



L3A: Higher-order Functions; Scope of Names

CS1101S: Programming Methodology

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26th August, 2026



Announcements

Nested declaration assignments

More fun with recursion

Higher-order functions

Scope of names



Announcements



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- ▶ Quests: they are “extra homework”! If you are struggling with your time management or work/life balance: Don’t touch the quests!



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- ▶ **Forum Hours.** Every Tuesday 6:30pm-8:30pm. Venue is COM1 basement.



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- ▶ Contest “Beautifully Built” opens today!
- ▶ **Forum Hours.** Every Tuesday 6:30pm-8:30pm. Venue is COM1 basement.
- ▶ Reading Assessment 1: Read our announcement carefully. Sample questions have been uploaded.



Some Words of Advice



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- ▶ Read the textbook



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- ▶ Use the substitution model (and stepper tool, if needed)



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- ▶ Think, then program



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- ▶ Think, then program
- ▶ Less is more



Some Words of Advice

- ▶ Read the textbook
- ▶ Use the substitution model (and stepper tool, if needed)
- ▶ Think, then program
- ▶ Less is more
- ▶ Use AI smartly (to optimise your learning)



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Names can refer to intermediate values

Example from SICPy 1.3.2

$$f(x, y) = x(1 + xy)^2 + y(1 - y) + (1 + xy)(1 - y)$$

Compute $f(2, 3)$

Names can refer to intermediate values

Example from SICPy 1.3.2

$$f(x, y) = x(1 + xy)^2 + y(1 - y) + (1 + xy)(1 - y)$$

Compute $f(2, 3)$

```
def f(x, y):  
    return (x * square(1 + x * y)  
            + y * (1 - y)  
            + (1 + x * y) * (1 - y))  
  
print(f(2, 3))
```


Names can refer to intermediate values

Example from SICPy 1.3.2

$$f(x, y) = x(1 + xy)^2 + y(1 - y) + (1 + xy)(1 - y)$$

Compute $f(2, 3)$

```
def f(x, y):  
    a = 1 + x * y  
    b = 1 - y  
    return x * square(a) + y * b + a * b  
  
print(f(2, 3))
```



Announcements

Nested declaration assignments

More fun with recursion

Example: fractal runes (“Rune Reading”)

Conditional statements (1.3.2)

Example: coin change (1.2.2)

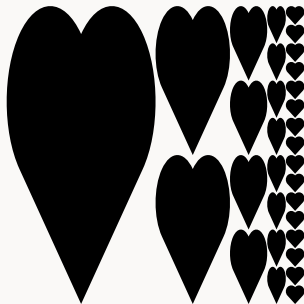
Higher-order functions

Scope of names



Example from Mission “Rune Reading”: `fractal`

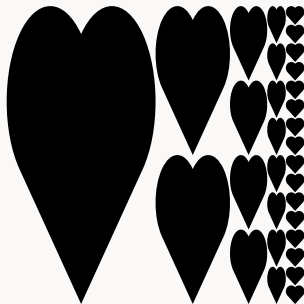
Define a function `fractal` that returns pictures like this:





Example from Mission “Rune Reading”: `fractal`

Define a function `fractal` that returns pictures like this:



when we call it like this: `fractal(heart, 5)`



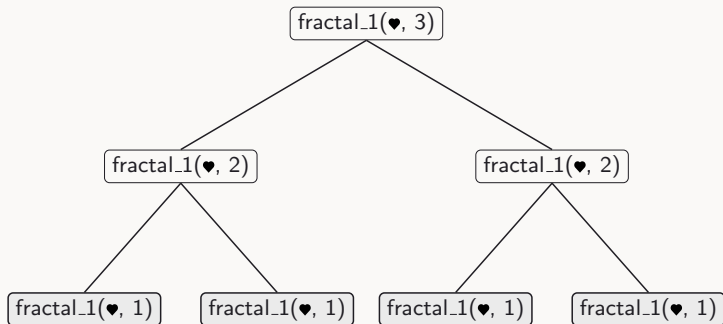
“Rune Reading” Example: fractal, Solution 1

```
def fractal_1(rune, n):  
    return (rune if n == 1  
            else beside(rune,  
                        stack(fractal_1(rune, n - 1),  
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```

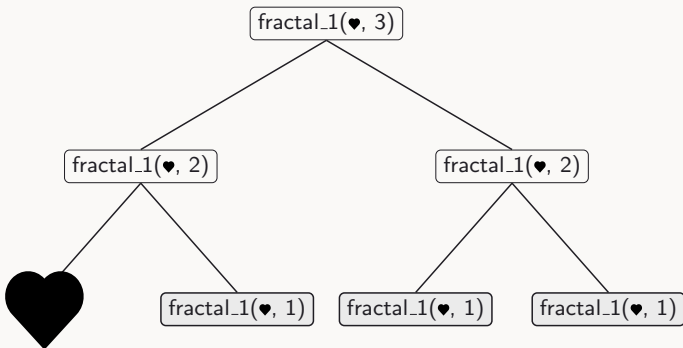




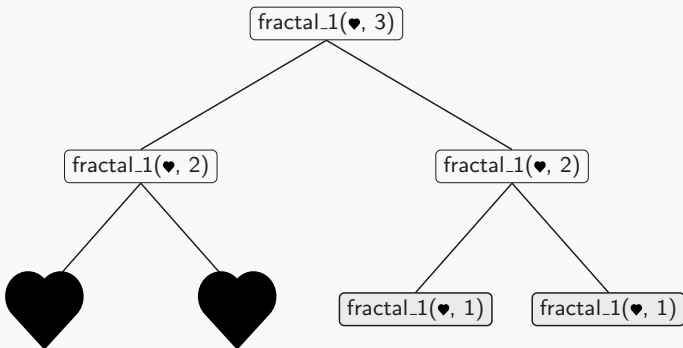
The Call Tree for `fractal_1(heart, 3)`



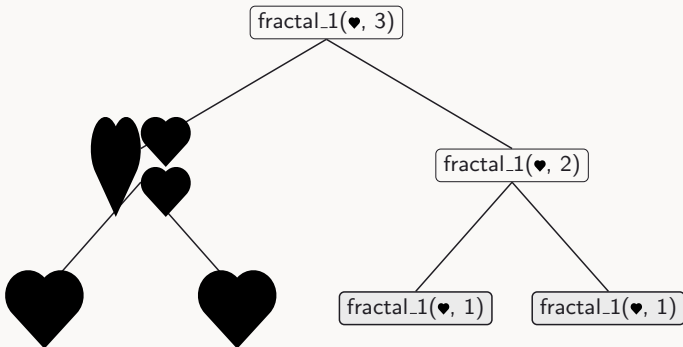
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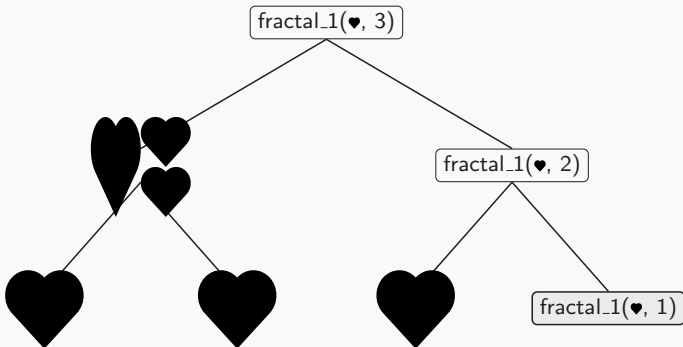
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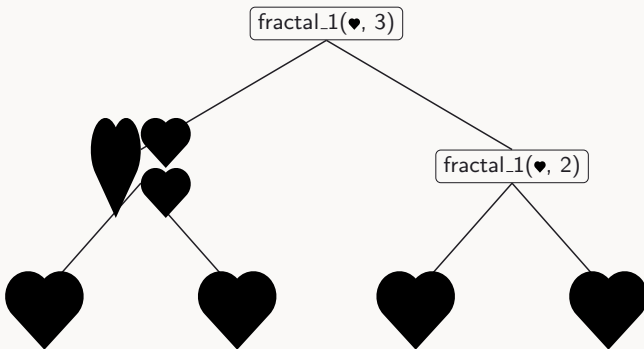
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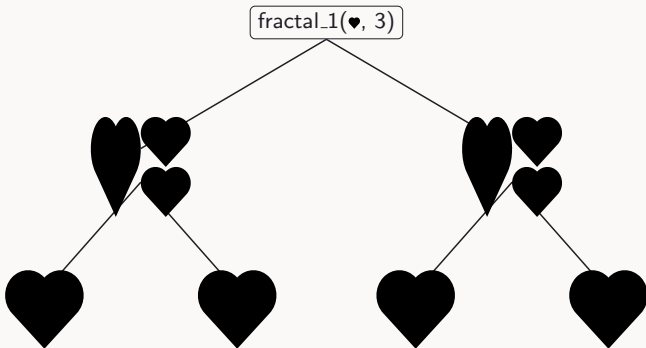
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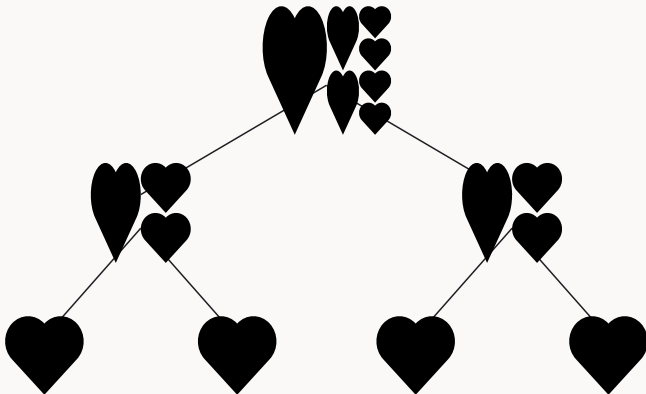
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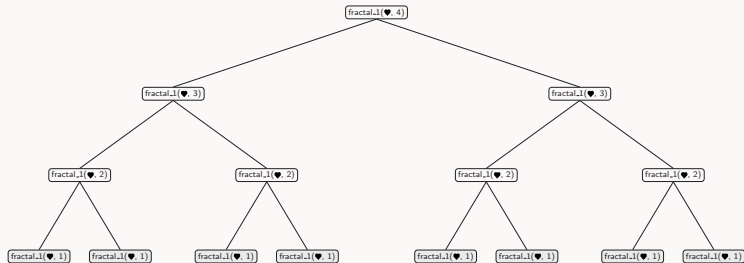


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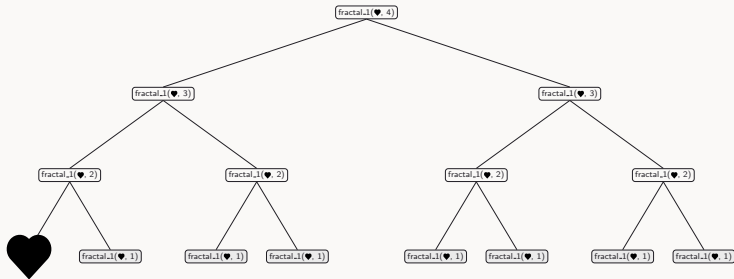


The Call Tree for `fractal_1(heart, 4)`



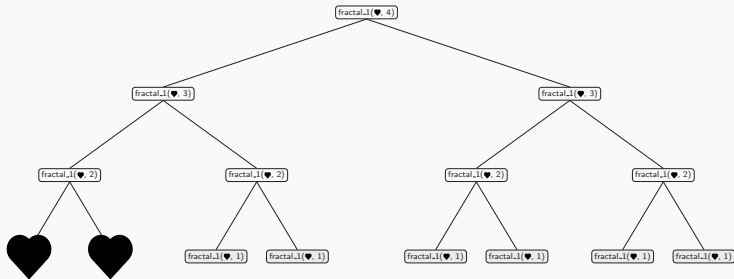


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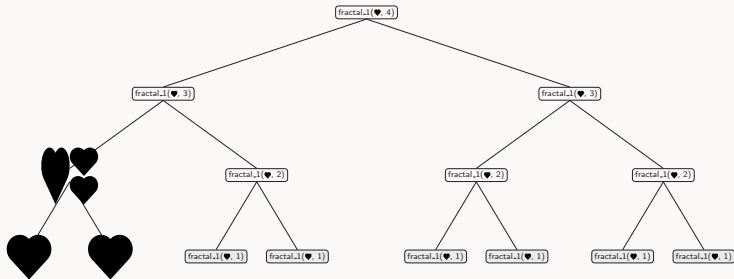


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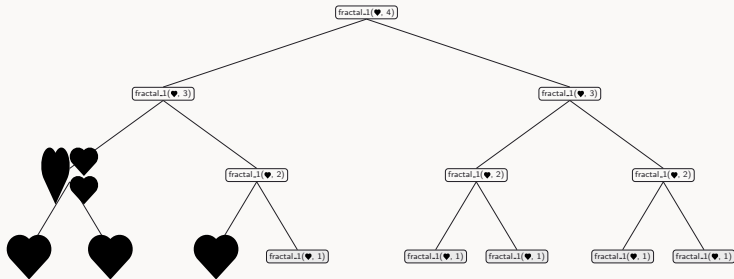


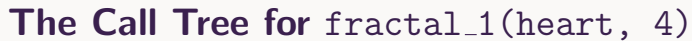
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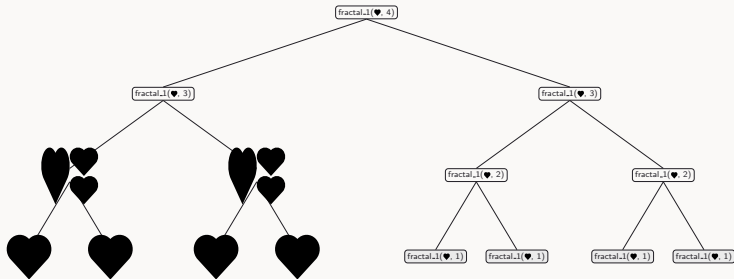
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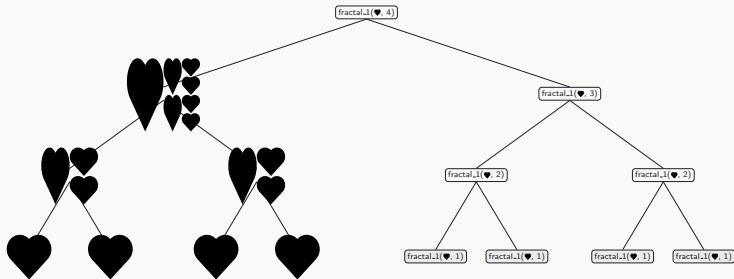


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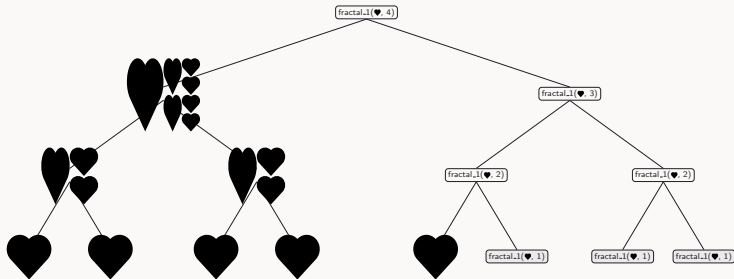


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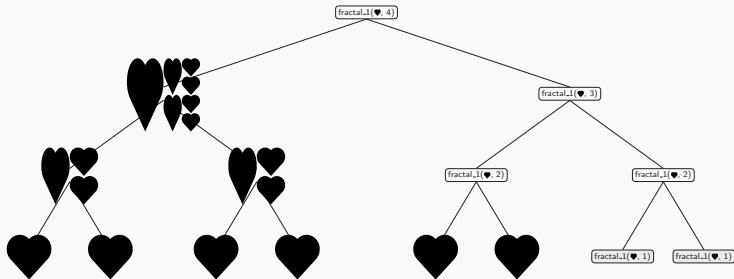


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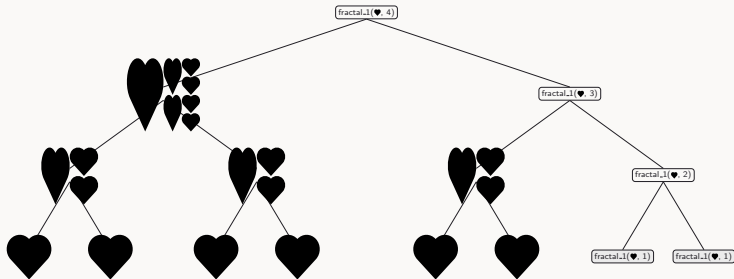


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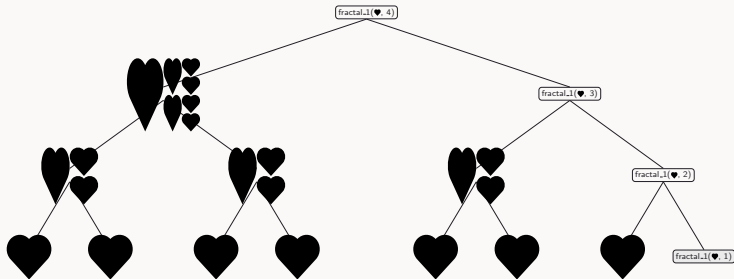


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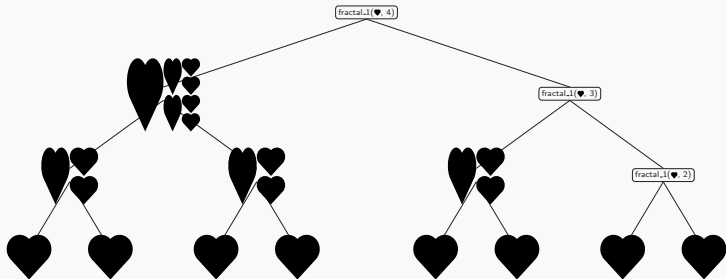


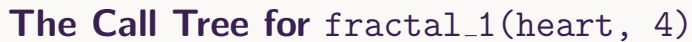
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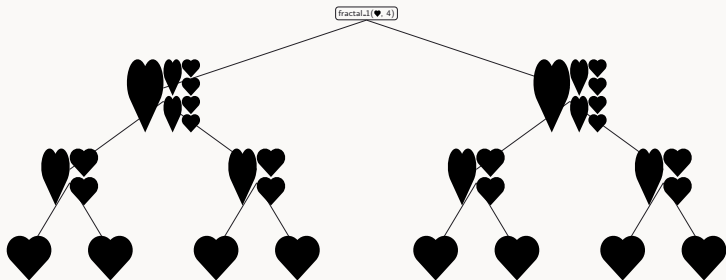


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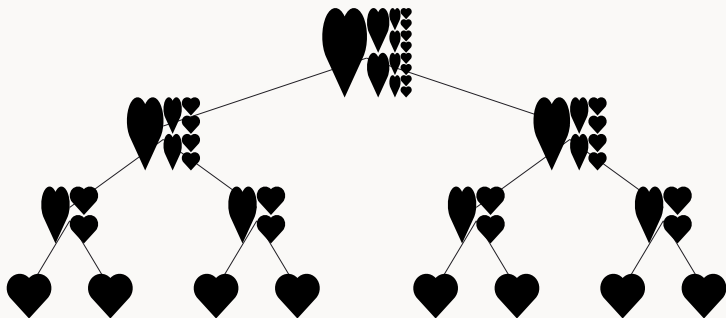




The Call Tree for `fractal_1(heart, 4)`



The Call Tree for `fractal_1(heart, 4)`





Can we avoid tree recursion?

```
def fractal_1(rune, n):  
    return (rune if n == 1  
            else beside(rune,  
                        stack(fractal_1(rune, n - 1),  
                              fractal_1(rune, n - 1))))
```



Can we avoid tree recursion?

```
def fractal_1(rune, n):  
    return (rune if n == 1  
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```



Question

Can we implement this function with linear recursion?



Is this a good idea?

```
def fractal_2(rune, n):  
    smaller_frac = fractal_2(rune, n - 1)  
    return (rune if n == 1  
            else beside(rune,  
                        stack(smaller_frac, smaller_frac)))
```





Is this a good idea?

```
def fractal_2(rune, n):  
    smaller_frac = fractal_2(rune, n - 1)  
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```



Can we declare a name...



Is this a good idea?

```
def fractal_2(rune, n):  
    smaller_frac = fractal_2(rune, n - 1)  
    return (rune if n == 1  
            else beside(rune,  
                        stack(smaller_frac, smaller_frac)))
```



Can we declare a name...

...just for the alternative of the conditional?



Conditional statements (see SICPy 1.3.2)

```
def fractal_3(rune, n):  
    if n == 1:  
        return rune  
    else:  
        f = fractal_3(rune, n - 1)  
        return beside(rune, stack(f, f))
```



Conditional statements (see SICPy 1.3.2)

```
def fractal_3(rune, n):  
    if n == 1:  
        return rune  
    else:  
        f = fractal_3(rune, n - 1)  
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```

- Each branch of the conditional can have local declarations.

Conditional statements (see SICPy 1.3.2)

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```

- ▶ Each branch of the conditional can have local declarations.
- ▶ A declaration assignment in a branch is only evaluated when the branch actually runs (more on scope shortly).

Conditional statements (see SICPy 1.3.2)

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        return beside(rune, stack(f, f))
```

- ▶ Each branch of the conditional can have local declarations.
- ▶ A declaration assignment in a branch is only evaluated when the branch actually runs (more on scope shortly).
- ▶ Remember to return a result *in each branch*. (Otherwise `None` is returned.)

λ

Example: coin change (1.2.2)

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- ▶ Given: Different kinds of coins (unlimited supply)

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- ▶ Given: Amount of money in cents

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- ▶ Wanted: Number of ways to change amount into coins



Example: coin change (1.2.2)

- ▶ Given: Different kinds of coins (unlimited supply)
- ▶ Given: Amount of money in cents
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Step 1

Read the problem *very carefully*



Play

- ▶ Given: Different kinds of coins (unlimited supply)
- ▶ Given: Amount of money in cents
- ▶ Wanted: Number of ways to change amount into coins



Play

- ▶ Given: Different kinds of coins (unlimited supply)
- ▶ Given: Amount of money in cents
- ▶ Wanted: Number of ways to change amount into coins
- ▶ How about 10 cents?



Play

- ▶ Given: Different kinds of coins (unlimited supply)
- ▶ Given: Amount of money in cents
- ▶ Wanted: Number of ways to change amount into coins
- ▶ How about 10 cents?
- ▶ How about 20 cents?



Play

- ▶ Given: Different kinds of coins (unlimited supply)
- ▶ Given: Amount of money in cents
- ▶ Wanted: Number of ways to change amount into coins
- ▶ How about 10 cents?
- ▶ How about 20 cents?
- ▶ 5 cents?



Play

- ▶ Given: Different kinds of coins (unlimited supply)
- ▶ Given: Amount of money in cents
- ▶ Wanted: Number of ways to change amount into coins
- ▶ How about 10 cents?
- ▶ How about 20 cents?
- ▶ 5 cents?
- ▶ 0 cents?



Play

- ▶ Given: Different kinds of coins (unlimited supply)
- ▶ Given: Amount of money in cents
- ▶ Wanted: Number of ways to change amount into coins
- ▶ How about 10 cents?
- ▶ How about 20 cents?
- ▶ 5 cents?
- ▶ 0 cents?
- ▶ -1 cents?



Play

- ▶ Given: Different kinds of coins (unlimited supply)
- ▶ Given: Amount of money in cents
- ▶ Wanted: Number of ways to change amount into coins
- ▶ How about 10 cents?
- ▶ How about 20 cents?
- ▶ 5 cents?
- ▶ 0 cents?
- ▶ -1 cents?

Step 2

Play with examples

λ

Think



Think

- ▶ Example: Number of ways to change 120 cents



Think

- ▶ Example: Number of ways to change 120 cents
- ▶ Idea: Highest coin is either 100 or not 100



Think

- ▶ Example: Number of ways to change 120 cents
- ▶ Idea: Highest coin is either 100 or not 100
- ▶ In the first case: Smaller problem



Think

- ▶ Example: Number of ways to change 120 cents
- ▶ Idea: Highest coin is either 100 or not 100
- ▶ In the first case: Smaller problem
- ▶ In the second case: Smaller problem



Think

- ▶ Example: Number of ways to change 120 cents
- ▶ Idea: Highest coin is either 100 or not 100
- ▶ In the first case: Smaller problem
- ▶ In the second case: Smaller problem
- ▶ Base cases?



Think

- ▶ Example: Number of ways to change 120 cents
- ▶ Idea: Highest coin is either 100 or not 100
- ▶ In the first case: Smaller problem
- ▶ In the second case: Smaller problem
- ▶ Base cases?

Step 3

Think: existing solution? divide-and-conquer? “wishful thinking”?



Representing the kinds of coins

```
def first_denomination(kinds_of_coins):  
    return (5 if kinds_of_coins == 1  
            else 10 if kinds_of_coins == 2  
            else 20 if kinds_of_coins == 3  
            else 50 if kinds_of_coins == 4  
            else 100 if kinds_of_coins == 5  
            else 0)
```





Idea in Python

```
def cc(amount, kinds_of_coins):  
    return (...) # base cases  
        if ...  
        else cc(amount -  
first_denomination(kinds_of_coins),  
                    kinds_of_coins)  
            + cc(amount, kinds_of_coins - 1))
```



Idea in Python

```
def cc(amount, kinds_of_coins):  
    return (...) # base cases  
        if ...  
        else cc(amount -  
first_denomination(kinds_of_coins),  
                    kinds_of_coins)  
            + cc(amount, kinds_of_coins - 1))
```

Step 4

Program using Python



Adding base cases

```
def cc(amount, kinds_of_coins):  
    return (1 if amount == 0  
            else 0 if amount < 0 or kinds_of_coins == 0  
            else cc(amount -  
                    first_denomination(kinds_of_coins), kinds_of_coins)  
                + cc(amount, kinds_of_coins - 1))
```



Adding base cases

```
def cc(amount, kinds_of_coins):  
    return (1 if amount == 0  
            else 0 if amount < 0 or kinds_of_coins == 0  
            else cc(amount -  
                    first_denomination(kinds_of_coins), kinds_of_coins)  
                + cc(amount, kinds_of_coins - 1))
```

Step 5

Test your program



The CS1101S 5-step Method of Problem Solving™

Step 1

Read the problem *very carefully*



The CS1101S 5-step Method of Problem Solving™

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Read the problem *very carefully*

Step 2

Play with examples



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Program using Python

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Announcements

Nested declaration assignments

More fun with recursion

Higher-order functions

- Functions as arguments (1.3.1)

- Lambda expressions (1.3.2)

- Functions as returned values

- Summary of new constructs today

Scope of names



Passing functions to functions

```
def f(g, x):  
    return g(x)  
  
def g(y):  
    return y + 1  
  
print(f(g, 7))
```





Passing more functions to functions

```
def f(g, x):  
    return g(g(x))  
  
def g(y):  
    return y + 1  
  
print(f(g, 7))
```





We have seen this before!

Repeating the pattern n times

```
print(repeat_apply(make_cross, 4, rcross))
```





We have seen this before!

Repeating the pattern n times

```
print(repeat_apply(make_cross, 4, rcross))
```



Combining and transforming pictures

```
def transform_mosaic(p1, p2, p3, p4, transform):  
    return transform(mosaic(p1, p2, p3, p4))
```




Look at this function (1.3.1)

```
def sum_integers(a, b):  
    return 0 if a > b else a + sum_integers(a + 1, b)
```





...and this one

```
def cube(x):  
    return x * x * x  
  
def sum_skip_cubes(a, b):  
    return 0 if a > b else cube(a) + sum_skip_cubes(a +  
2, b)
```





Abstraction (1.3.1)

```
def sum(a, b):  
    return (0 if a > b  
           else ⟨compute value with a⟩  
           + sum(⟨next value from a⟩, b))
```



Abstraction (1.3.1)

```
def sum(a, b):  
    return (0 if a > b  
           else ⟨compute value with a⟩  
           + sum(⟨next value from a⟩, b))
```

in Python:

```
def sum(term, a, next, b):  
    return 0 if a > b else term(a) + sum(term, next(a),  
    next, b)
```



sum_integers using sum

```
def identity(x):  
    return x  
  
def plus_one(x):  
    return x + 1  
  
def sum_integers(a, b):  
    return sum(identity, a, plus_one, b)
```



sum_skip_cubes **using** sum

```
def cube(x):  
    return x * x * x  
  
def plus_two(x):  
    return x + 2  
  
def sum_skip_cubes(a, b):  
    return sum(cube, a, plus_two, b)
```



sum_skip_cubes **using** sum

```
def cube(x):  
    return x * x * x  
  
def plus_two(x):  
    return x + 2  
  
def sum_cubes(a, b):  
    return sum(cube, a, plus_two, b)
```



Visibility



sum_skip_cubes **using** sum

```
def cube(x):  
    return x * x * x  
  
def plus_two(x):  
    return x + 2  
  
def sum_cubes(a, b):  
    return sum(cube, a, plus_two, b)
```



Visibility

Can we “hide” `cube` and `plus_two` inside of `sum_skip_cubes`?



Yes, we can! (see also SICPy 1.1.8)

```
def sum_skip_cubes(a, b):  
    def cube(x):  
        return x * x * x  
    def plus_two(x):  
        return x + 2  
    return sum(cube, a, plus_two, b)
```



λ

Now You See It, Now You Don't



Now You See It, Now You Don't

Let's tuck `fact_iter` in and see how many parameters it can shed along the way!



Now You See It, Now You Don't

Let's tuck `fact_iter` in and see how many parameters it can shed along the way!

Look for Check-In I3A in the Source Academy!



Another look at such local functions

```
def sum_skip_cubes(a, b):  
    def cube(x):  
        return x * x * x  
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```





Another look at such local functions

```
def sum_skip_cubes(a, b):  
    def cube(x):  
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```



This is still quite verbose



Another look at such local functions

```
def sum_skip_cubes(a, b):  
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```



This is still quite verbose

Do we need all these words such as `def`, `return`?



Another look at such local functions

```
def sum_skip_cubes(a, b):  
    def cube(x):  
        return x * x * x  
    def plus_two(x):  
        return x + 2  
    return sum(cube, a, plus_two, b)
```



This is still quite verbose

Do we need all these words such as `def`, `return`?

Do we need to even give names to these functions?



No, we don't! Lambda expressions

```
def sum_skip_cubes(a, b):  
    def cube(x):  
        return x * x * x  
    def plus_two(x):  
        return x + 2  
    return sum(cube, a, plus_two, b)
```

```
# instead just write:  
def sum_skip_cubes(a, b):  
    return sum(lambda x: x * x * x, a, lambda x: x + 2, b)
```



Lambda expressions (1.3.2)

New kind of expression

`lambda parameters : expression`



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`lambda parameters : expression`

Meaning

The expression evaluates to a function value.

Function has given *parameters* and *expression* as its (only) return value; no `return` keyword needed.



Lambda expressions (1.3.2)

New kind of expression

`lambda parameters : expression`

Meaning

The expression evaluates to a function value.

Function has given *parameters* and *expression* as its (only) return value; no `return` keyword needed.



(yes, this is also why it's on our t-shirts)



We can have more than one parameter

Comma-separated, just like a function definition

`lambda  $p_1$ ,  $p_2$ , \dots,  $p_n$  : expression`



We can have more than one parameter

Comma-separated, just like a function definition

$\text{lambda } p_1, p_2, \dots, p_n : \text{expression}$

```
add_three = lambda x, y, z: x + y + z  
  
print(add_three(1, 2, 3))
```





An alternative syntax for function definition

```
def plus4(x):  
    return x + 4
```



can be written as

```
plus4 = lambda x: x + 4
```





Returning Functions from Functions

```
def make_adder(x):  
    def adder(y):  
        return x + y  
    return adder  
  
adder_four = make_adder(4)  
print(adder_four(6))
```





...or with the new lambda expressions

```
def make_adder(x):  
    return lambda y: x + y  
  
adder_four = make_adder(4)  
print(adder_four(6))
```





Returning Functions from Functions

```
def make_adder(x):  
    return lambda y: x + y  
  
print((make_adder(4))(6))  
  
# we can drop the parentheses:  
#  
# make_adder(4)(6)
```





Returning Functions from Functions

```
def make_adder(x):  
    return lambda y: x + y  
  
adder_1 = make_adder(1)  
adder_2 = make_adder(2)  
  
# adder_1(10) returns 11  
print(adder_1(10))  
  
# adder_2(20) returns 22  
print(adder_2(20))
```





Summary of new constructs today

- ▶ Nested declaration assignment and function definition statements
- ▶ Conditional statements
- ▶ Lambda expressions



Announcements

Nested declaration assignments

More fun with recursion

Higher-order functions

Scope of names

- Examples

- Overview of scoping rules

- The details



Scope of names: an example

```
z = 2
def f(g):
    z = 4
    return g(z)

print(f(lambda y: y + z))
```



Scope of names: an example

```
z = 2
def f(g):
    z = 4
    return g(z)

print(f(lambda y: y + z))
```



Questions about scope

What names are declared by this program?



Scope of names: an example

```
z = 2
def f(g):
    z = 4
    return g(z)

print(f(lambda y: y + z))
```



Questions about scope

What names are declared by this program?

Which declaration does each name occurrence refer to?



Scope of names: another example

```
x = 10
def square(x):
    return x * x

def addx(y):
    return y + x

print(square(x + 5) * addx(x + 20))
```



Questions about scope

Which declaration does each occurrence of `x` refer to?



Scope of names: yet another example

```
pi = 3.141592653589793

def circle_area_from_radius(r):
    pi = 22 / 7
    return pi * square(r)
```



Questions about scope

Which declaration does the occurrence of `pi` refer to?



Scope of names: hypotenuse example

```
def square(x):  
    return x * x  
  
def hypotenuse(a, b):  
    def sum_of_squares():  
        return square(a) + square(b)  
    return math.sqrt(sum_of_squares())
```



Scope of names: hypotenuse example

```
def square(x):  
    return x * x  
  
def hypotenuse(a, b):  
    def sum_of_squares():  
        return square(a) + square(b)  
    return math_sqrt(sum_of_squares())
```

Names can refer to declarations outside of the immediately surrounding function definition.



Overview of scoping rules

Declarations mandatory

All names in Python must be declared.



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Forms of declaration

1. Primitive names
2. Declaration assignments
3. Parameters of function definitions and lambda expressions
4. Function name of function definitions



Overview of scoping rules

Declarations mandatory

All names in Python must be declared.

Forms of declaration

1. Primitive names
2. Declaration assignments
3. Parameters of function definitions and lambda expressions
4. Function name of function definitions

Scoping rule

A name occurrence refers to the *closest surrounding declaration* of the name.



Forms of Declaration

(1) Primitive names

The [Python §1](#) pages tell us what names are primitive, e.g.
`math.floor`.



Forms of Declaration

(1) Primitive names

The [Python §1](#) pages tell us what names are primitive, e.g. `math.floor`.

We can also `import` further primitive names from modules. For example, from the `rune` module:

```
from rune import heart, quarter_turn_right
```



Forms of Declaration

(2) Declaration assignments

The scope of a declaration assignment is the body of the *immediately surrounding function*, or the whole program, if there is none.

A simple example (see [SICPy 1.3.2](#))

```
z = 3
def f():
    print(z)
    z = 4
f()
```





Forms of Declaration

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A name declared in a function cannot be used before the declaration is fully evaluated, regardless of whether the same name is declared outside the function.



Forms of Declaration

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```
z = 3
def f():
    print(z)
    z = 4
f()
```



A name declared in a function cannot be used before the declaration is fully evaluated, regardless of whether the same name is declared outside the function. Here, `print(z)` runs before the declaration `z = 4`.



Forms of Declaration

(2) Declaration assignments: what about branches?

```
z = 100
def f(x, y):
    if x > 0:
        z = x * y
    return z
print(f(3, 4))
```





Forms of Declaration

(2) Declaration assignments: what about branches?

```
z = 100
def f(x, y):
    if x > 0:
        z = x * y
    return z
print(f(3, 4))
```



The declaration assignment `z = x * y` is inside the `if`, but the local `z` still shadows the global one throughout `f`.



Forms of Declaration

(2) Declaration assignments: scope vs. value

z 's *scope* covers all of f , but it only has a *value* once the assignment has actually run.

```
z = 100

def f(x, y):
    if x > 0:
        z = x * y
    return z

print(f(-3, 4))
```





Forms of Declaration

(2) Declaration assignments: scope vs. value

z 's *scope* covers all of f , but it only has a *value* once the assignment has actually run.

```
z = 100

def f(x, y):
    if x > 0:
        z = x * y
    return z

print(f(-3, 4))
```

$f(-3, 4)$ skips the `if`: z is still in scope but never gets a value (*not even the global z above*), so this is an error.



Forms of Declaration

(3) Parameters

The scope of the parameters of a lambda expression or function definition is the body of the function.

```
def f(x, y, z):  
    ... x ... y ... z ...  
  
lambda v, w, u: ... v ... w ... u ...
```



Forms of Declaration

(4) Function name

The scope of the function name of a function definition is as if the function was declared with a declaration assignment.

```
def f(x):  
    ...
```

as if we wrote

```
f = ...
```



Lexical scoping

Scoping rule

A name occurrence refers to the *closest surrounding declaration* of the name.

```
def f(x):  
    def g(w):  
        x = ...  
        y = x + 1  
        return lambda x: ...x...  
    return g(x)
```



Important Ideas



Important Ideas

- ▶ *Hiding* can be a useful abstraction technique



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- ▶ *Recursion* is an elegant pattern of problem solving



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Important Ideas

- ▶ *Hiding* can be a useful abstraction technique
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- ▶ Functions can be *passed to* functions
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- ▶ Higher-order functions are *useful* for building abstractions
- ▶ With nested functions, we need to understand the *scope* of names
- ▶ The CS1101S 5-step Method of Problem Solving™



Concluding Unit 1



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- ▶ Friday: Curves and Sound
- ▶ Look out for Unit 2 with Prof Sanka: Building Abstractions with Data, starting with L4